

Ex. No: 8 Deadlock Detection Algorithm

PROGRAM: DEADLOCK DETECTION ALGORITHM

C PROGRAM

```
#include <stdio.h>

int main()
{
    int allocation[10][10], request[10][10];
    int available[10];
    int work[10];
    int finish[10];
    int n, m;
    int i, j, k;
    int found;

    printf("Enter Number of Processes: ");
    scanf("%d", &n);
    printf("Enter Number of Resource Types: ");
    scanf("%d", &m);
    printf("\nEnter Allocation Matrix:
\n");

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < m; j++)
        {
            scanf("%d", &allocation[i][j]);
        }
    }

    printf("\nEnter Request Matrix:
```

```
\n");
```

```
for(i = 0; i < n; i++)
```

```
{
```

```
for(j = 0; j < m; j++)
```

```
{
```

```
scanf("%d", &request[i][j]);
```

```
}
```

```
}
```

```
printf("
```

```
\nEnter Available Resources:
```

```
\n");
```

```
for(i = 0; i < m; i++)
```

```
{
```

```
scanf("%d", &available[i]);
```

```
work[i] = available[i];
```

```
}
```

```
for(i = 0; i < n; i++)
```

```
{
```

```
finish[i] = 0;
```

```
}
```

```
do
```

```
{
```

```
found = 0;
```

```
for(i = 0; i < n; i++)
```

```
{
```

```
if(finish[i] == 0)
```

```
{
```

```
for(j = 0; j < m; j++)
```

```
{
```

```

if(request[i][j] > work[j])
break;
}
if(j == m)
{
for(k = 0; k < m; k++)

{

work[k] += allocation[i][k];

}

finish[i] = 1;
found = 1;
}
}
}
} while(found);
found = 0;
printf(
"
\nDeadlocked Processes:
\n");

for(i = 0; i < n; i++)
{
if(finish[i] == 0)
{
printf("P%d ", i);
found = 1;

```

```

}
}
if(found == 0)
{
printf("No Deadlock Detected");
}
printf("
\n");
return 0;
}

```

Output:

```

(kali㉿kali)-[~]
└─$ ./deadlock
Enter Number of Processes: 2
Enter Number of Resource Types: 2
Enter Allocation Matrix:
4
3
2
3
Enter Request Matrix:
6
5
4
3
Enter Available Resources:
2
3
Deadlocked Processes:
P0 P1

```

SHELL SCRIPT

```

#!/bin/bash

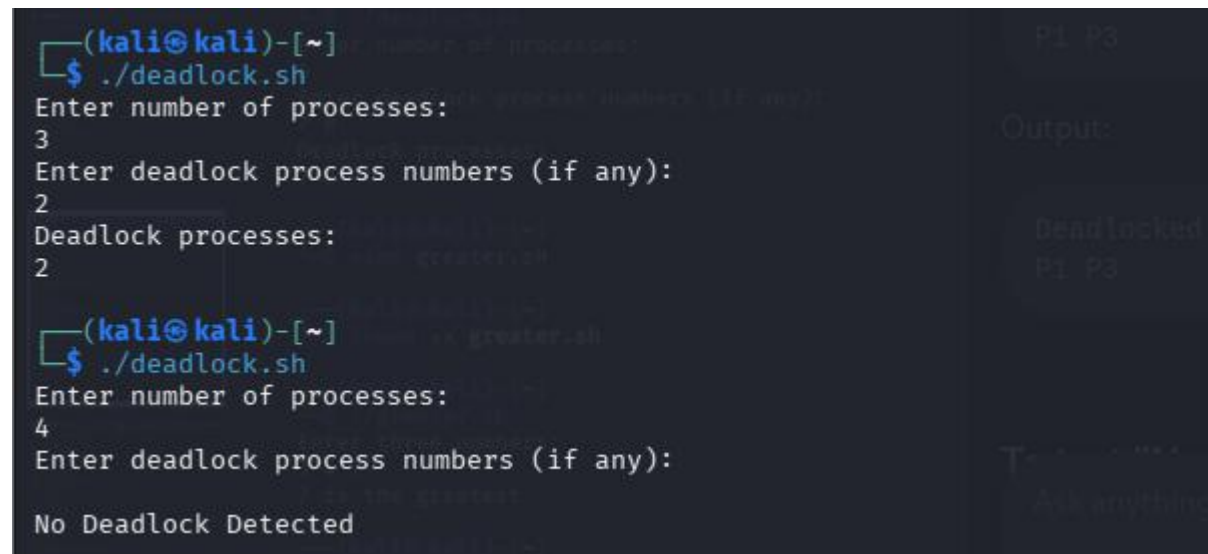
echo "Enter number of processes:"
read n

echo "Enter deadlocked process numbers (if any):"
read processes

```

```
if [ -z "$processes" ]  
then  
echo "No Deadlock Detected"  
else  
echo "Deadlocked Processes:"  
echo "$processes"  
fi
```

Output:



The screenshot shows two separate executions of the script `./deadlock.sh` in a Kali Linux terminal. In the first execution, the user enters 3 for the number of processes and 2 for the deadlock process numbers. The script outputs "Deadlocked Processes:" followed by "P1 P3". In the second execution, the user enters 4 for the number of processes and no deadlock process numbers. The script outputs "No Deadlock Detected".

```
(kali㉿kali)-[~]  
$ ./deadlock.sh  
Enter number of processes:  
3  
Enter deadlock process numbers (if any):  
2  
Deadlock processes:  
2  
P1 P3  
  
(kali㉿kali)-[~]  
$ ./deadlock.sh  
Enter number of processes:  
4  
Enter deadlock process numbers (if any):  
No Deadlock Detected
```